SZABIST
University

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ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

Total Marks:_____

Obtained Marks: _____

Computing Vision

Lab Task # 01

Submitted To: **Mr. Muhammad Azhar**

Student Name: **Ayesha Munir**

Reg. Number: **23108106**

1) List at least 10 features / functional requirements of your project

Prepare a scenario/case study for your own project idea and provide feature list relevant to the proposed scenario.

ADD KEYWORDS FOR SPECIFIC SUBJECT AND PROJECT

2) Provide a feasibility study

Prepare a table mapping each requirement to available Tools/resources/APIs/packages/libraries that would help you implement.

Sr.	Requirements	Resources

3) Technical developer guide

Choose any three features from your feature list and provide a detailed, step-by-step technical guide on how to implement them.

E.g. Step by step procedure to install a required library, to configure an API, and to implement a requirement by using specific Tools/resources/APIs/packages/libraries.

4) Implementation

Using any ONE of the available tools/technology stack, implement the three selected features inside a small working project.

Submission Requirements:

- Push your completed mini project to your own GitHub repository.
- Include a short README file describing the added features and navigation flow.
- Prepare screenshots of your project code from android studio, output from emulator or separate device, and your git repository along with feasibility study to include in your submission document.

Solution:

Real-Time Motion Detection & Intrusion Alert System

Scenario / Case Study

A small office wants a low-cost surveillance system that detects motion after office hours and logs intrusion events automatically.

The system uses a webcam to detect movement and records time-stamped alerts.

Keywords

Computer Vision, Motion Detection, Frame Differencing, OpenCV, Image Processing, Thresholding, Contour Detection, Security Monitoring, Real-Time Processing

Functional Requirements

1. Capture live webcam feed
2. Convert frames to grayscale
3. Apply Gaussian blur
4. Perform frame differencing
5. Apply thresholding
6. Detect contours
7. Identify motion area
8. Display motion alert message
9. Log motion event with date & time
10. Store logs in text file
11. Exit using ESC key

Feasibility Study

Requirement Mapping

Sr	Requirement	Tools / Libraries
1	Webcam capture	OpenCV
2	Frame differencing	absdiff()
3	Image preprocessing	cvtColor(), GaussianBlur()
4	Thresholding	threshold()
5	Contour detection	findContours()
6	Motion detection	Area comparison
7	Logging	Python file handling
8	Timestamp	datetime module
9	Display alert	putText()
10	Exit control	waitKey()

Technical Feasibility

Uses OpenCV which supports motion detection functions natively.

Economic Feasibility

Fully open-source. No paid services required.

Operational Feasibility

Can run on standard PC with webcam. Easy to deploy in small offices.

Technical Developer Guide

Selected Features:

1. Frame Differencing
2. Contour Detection
3. Motion Logging

Feature 1: Frame Differencing

```
frame_diff = cv2.absdiff(first_frame, gray)
```

Feature 2: Contour Detection

```
contours, _ = cv2.findContours(thresh.copy(),
                               cv2.RETR_EXTERNAL,
                               cv2.CHAIN_APPROX_SIMPLE)
```

Feature 3: Motion Logging

with open("motion_log.txt","a") as f:

f.write("Motion detected at " + time + "\n")

```
import cv2
from datetime import datetime

cap = cv2.VideoCapture(0)
first_frame = None

while True:
    ret, frame = cap.read()
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    gray = cv2.GaussianBlur(gray,(21,21),0)

    if first_frame is None:
        first_frame = gray
        continue

    frame_diff = cv2.absdiff(first_frame, gray)
    thresh = cv2.threshold(frame_diff,25,255,cv2.THRESH_BINARY)[1]
    contours,_ = cv2.findContours(thresh.copy(),
                                  cv2.RETR_EXTERNAL,
                                  cv2.CHAIN_APPROX_SIMPLE)

    for contour in contours:
        if cv2.contourArea(contour) < 1000:
            continue
```

```
        (x,y,w,h) = cv2.boundingRect(contour)
        cv2.rectangle(frame,(x,y),(x+w,y+h),(0,0,255),2)

        now = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
        with open("motion_log.txt","a") as f:
            f.write("Motion detected at "+now+"\n")

    cv2.imshow("Motion Detection",frame)

    if cv2.waitKey(1)==27:
        break

cap.release()
cv2.destroyAllWindows()
```

MotionDetectionSystem Public

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23108106-sketch Add files via upload 9efbf6c · 1 minute ago 3 Commits

README.md	Update README.md	1 minute ago
app.py	Add files via upload	1 minute ago
motion_log.txt	Add files via upload	1 minute ago

README

Motion Detection & Intrusion Alert System

Description

This project implements a real-time motion detection system using OpenCV. It detects movement via frame differencing and logs intrusion events.

Features

- Real-time motion detection
- Frame differencing technique
- Contour detection
- Motion alert display
- Event logging with timestamp
- Text-based log storage

Technologies Used

- Python
- OpenCV
- datetime module

How to Run

pip install opencv-python python app.py

Output

- Webcam window with bounding box
- motion_log.txt file storing alerts